EX NO : 7 DATE : 15-07-26	PERFORMANCE METRIC FOR CLASSIFIERS

AIM:

To build and evaluate a Decision Tree Classifier with and without pre-pruning using a synthetic dataset.

PROBLEM STATEMENT:

A synthetic dataset with 1000 student records containing study hours, attendance, assignment marks, internal marks, and result (Pass/Fail). The first four attributes are input features, while result is the target variable for classification.

TASKS:

Implement the following in Python
1. Create a synthetic dataset with columns study-hours, attd, assign-marks, internal marks, result(pass/fail) with 1000 rows using code
2. Build a decision tree classifier on this without any restrcitions
3. Pre-Prune the tree using any one logic of pre-pruning
4. Display all the five performance metrics along with confusion/classification matrix

CODE:

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    confusion_matrix,
    classification_report
)

np.random.seed(42)

n = 1000

study = np.random.randint(1, 11, n)
att = np.random.randint(50, 101, n)
assign = np.random.randint(0, 21, n)
internal = np.random.randint(0, 51, n)

score = (
    study * 3 +
    att * 0.3 +
    assign * 2 +
    internal * 1.5
)

result = np.where(score >= 90, "Pass", "Fail")

df = pd.DataFrame({
```

```
    "study": study,
    "att": att,
    "assign": assign,
    "internal": internal,
    "result": result
})

print("Dataset Sample")
print(df.head())

df["result"] = df["result"].map({"Fail": 0, "Pass": 1})

X = df.drop("result", axis=1)
y = df["result"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

full_tree = DecisionTreeClassifier(random_state=42)

full_tree.fit(X_train, y_train)

pred_full = full_tree.predict(X_test)

print("\nUNRESTRICTED DECISION TREE")
print("Accuracy :", accuracy_score(y_test, pred_full))
print("Precision:", precision_score(y_test, pred_full))
print("Recall   :", recall_score(y_test, pred_full))
print("F1 Score :", f1_score(y_test, pred_full))

cm = confusion_matrix(y_test, pred_full)

tn, fp, fn, tp = cm.ravel()

spec = tn / (tn + fp)

print("Specificity:", spec)

print("\nConfusion Matrix")
print(cm)

print("\nClassification Report")
print(classification_report(y_test, pred_full))

pruned_tree = DecisionTreeClassifier(
    max_depth=4,
    random_state=42
)

pruned_tree.fit(X_train, y_train)

pred_pruned = pruned_tree.predict(X_test)

print("\nPRE-PRUNED DECISION TREE (max_depth=4)")
print("Accuracy :", accuracy_score(y_test, pred_pruned))
print("Precision:", precision_score(y_test, pred_pruned))
print("Recall   :", recall_score(y_test, pred_pruned))
print("F1 Score :", f1_score(y_test, pred_pruned))
```

```
cm = confusion_matrix(y_test, pred_pruned)

tn, fp, fn, tp = cm.ravel()

spec = tn / (tn + fp)

print("Specificity:", spec)

print("\nConfusion Matrix")
print(cm)

print("\nClassification Report")
print(classification_report(y_test, pred_pruned))
```

OUTPUT:

```
Dataset Sample
   study  att  assign  internal result
0      7   82      16         7   Fail
1      4   96      18        22   Pass
2      8   89      13         2   Fail
3      5   92      11         3   Fail
4      7   61       9        24   Pass

UNRESTRICTED DECISION TREE
Accuracy : 0.905
Precision: 0.9230769230769231
Recall   : 0.897196261682243
F1 Score : 0.909952606635071
Specificity: 0.9139784946236559

Confusion Matrix
[[85  8]
 [11 96]]

Classification Report
              precision    recall  f1-score   support

           0       0.89      0.91      0.90        93
           1       0.92      0.90      0.91       107

    accuracy                           0.91       200
   macro avg       0.90      0.91      0.90       200
weighted avg       0.91      0.91      0.91       200
```

```
PRE-PRUNED DECISION TREE (max_depth=4)
Accuracy : 0.885
Precision: 0.8962264150943396
Recall   : 0.8878504672897196
F1 Score : 0.892018779342723
Specificity: 0.8817204301075269

Confusion Matrix
[[82 11]
 [12 95]]

Classification Report
              precision    recall  f1-score   support

           0       0.87      0.88      0.88        93
           1       0.90      0.89      0.89       107

    accuracy                           0.89       200
   macro avg       0.88      0.88      0.88       200
weighted avg       0.89      0.89      0.89       200
```